

Kubernetes Quiz

Kubernetes Quiz Questions

1. What is the smallest and simplest deployable unit object created and managed by Kubernetes?

- ☐ A. Node
- ☐ B. Container
- ☐ C. Pod
- ☐ D. Deployment

Correct Answer: C

Explanation: A Pod represents a single instance of a running process in a cluster and can contain one or more containers. Nodes host Pods.

Domain: Kubernetes Fundamentals

Competency: Kubernetes Resources

2. Which Kubernetes component is responsible for watching for newly created Pods and assigning them to Nodes?

- ☐ A. kube-apiserver
- ☐ B. etcd
- ☐ C. kube-scheduler
- ☐ D. kubelet

Correct Answer: C

Explanation: The kube-scheduler is the control plane component that decides which Node a Pod should run on based on constraints and availability.

Domain: Kubernetes Fundamentals

Competency: Kubernetes Architecture

3. What is the primary function of the kubelet component?

- ☐ A. Storing cluster state
- ☐ B. Scheduling Pods onto Nodes
- ☐ C. Managing the container runtime
- ☐ D. Exposing the Kubernetes API

Correct Answer: C

Explanation: The kubelet runs on each Node and ensures that containers described in PodSpecs are running and healthy. It interacts with the container runtime.

Domain: Kubernetes Fundamentals

Competency: Kubernetes Architecture

4. Which component acts as the central control plane and exposes the Kubernetes API?

- ☐ A. kube-proxy
- ☐ B. kube-apiserver
- ☐ C. etcd
- ☐ D. controller-manager

Correct Answer: B

Explanation: The kube-apiserver validates and configures data for API objects (Pods, Services, etc.) and is the frontend for the control plane.

Domain: Kubernetes Fundamentals

Competency: Kubernetes API

5. What is the primary role of etcd in a Kubernetes cluster?

- ☐ A. Running application containers
- ☐ B. Scheduling workloads
- ☐ C. Storing the cluster state
- ☐ D. Managing network policies

Correct Answer: C

Explanation: etcd is a consistent and highly-available key-value store used as Kubernetes' backing store for all cluster data.

Domain: Kubernetes Fundamentals

Competency: Kubernetes Architecture

6. Which Kubernetes resource is typically used to manage stateless applications by ensuring a specified number of Pod replicas are running?

- ☐ A. StatefulSet
- ☐ B. DaemonSet
- ☐ C. Deployment
- ☐ D. Job

Correct Answer: C

Explanation: Deployments manage ReplicaSets, providing declarative updates and scaling for stateless applications.

Domain: Kubernetes Fundamentals

Competency: Kubernetes Resources

7. What type of software package bundles application code with all its dependencies, libraries, and configuration files?

- ☐ A. Virtual Machine
- ☐ B. Container
- ☐ C. Operating System
- ☐ D. Serverless Function

Correct Answer: B

Explanation: Containers package application code and dependencies together, isolating them from the underlying infrastructure.

Domain: Kubernetes Fundamentals

Competency: Containers

8. In Kubernetes, what is the process of assigning Pods to Nodes called?

- ☐ A. Replication
- ☐ B. Orchestration
- ☐ C. Scheduling
- ☐ D. Deployment

Correct Answer: C

Explanation: Scheduling is the core function of the kube-scheduler, selecting appropriate Nodes for Pods based on resource requests and other factors.

Domain: Kubernetes Fundamentals

Competency: Scheduling

9. What is the main benefit of using container orchestration systems like Kubernetes?

- ☐ A. Simplifying code writing
- ☐ B. Automating deployment & scaling
- ☐ C. Reducing storage costs
- ☐ D. Improving network latency

Correct Answer: B

Explanation: Orchestrators automate the deployment, scaling, management, and networking of containerized applications.

Domain: Container Orchestration

Competency: Container Orchestration Fundamentals

10. Which of these is a standard specified by the OCI (Open Container Initiative)?

- ☐ A. Docker
- ☐ B. Kubernetes

- ☐ C. Image Specification
- ☐ D. Istio

Correct Answer: C

Explanation: The OCI defines standards for container image formats (Image Specification) and runtime (Runtime Specification).

Domain: Kubernetes Fundamentals

Competency: Containers

11. Which container runtime is most commonly associated with Kubernetes and implements the CRI (Container Runtime Interface)?

- ☐ A. Docker (dockershim)
- ☐ B. containerd
- ☐ C. rkt
- ☐ D. LXC

Correct Answer: B

Explanation: containerd is a widely used container runtime focused on simplicity, robustness, and portability, fully implementing the CRI.

Domain: Container Orchestration

Competency: Runtime

12. What Kubernetes object provides a stable IP address and DNS name for accessing a set of Pods?

- ☐ A. Ingress
- ☐ B. Service
- ☐ C. EndpointSlice
- ☐ D. NetworkPolicy

Correct Answer: B

Explanation: A Service defines a logical set of Pods and a policy by which to access them, providing load balancing and service discovery.

Domain: Container Orchestration

Competency: Networking

13. How can you securely store sensitive information like passwords or API keys in Kubernetes?

- ☐ A. ConfigMap
- ☐ B. Annotation
- ☐ C. Label
- ☐ D. Secret

Correct Answer: D

Explanation: Secrets are Kubernetes objects specifically designed to hold small amounts of sensitive data.

Domain: Container Orchestration

Competency: Security

14. What is the role of a CNI (Container Network Interface) plugin in Kubernetes?

- ☐ A. Managing storage volumes
- ☐ B. Providing container networking
- ☐ C. Scheduling Pods
- ☐ D. Securing the API server

Correct Answer: B

Explanation: CNI plugins are responsible for configuring network interfaces for containers and managing IP address allocation within the cluster.

Domain: Container Orchestration

Competency: Networking

15. Which Kubernetes object defines rules about how Pods are allowed to communicate with each other and network endpoints?

- ☐ A. Service
- ☐ B. Ingress
- ☐ C. NetworkPolicy
- ☐ D. SecurityContext

Correct Answer: C

Explanation: NetworkPolicies allow specifying traffic flow rules at the IP address or port level (OSI layer 3 or 4).

Domain: Container Orchestration

Competency: Networking

16. What is the primary purpose of a Service Mesh like Istio or Linkerd?

- ☐ A. Container image building
- ☐ B. Cluster storage management
- ☐ C. Managing inter-service comms
- ☐ D. Node provisioning

Correct Answer: C

Explanation: Service meshes add observability, security, and reliability features to communication between microservices (service-to-service).

Domain: Container Orchestration

Competency: Service Mesh

17. What Kubernetes object represents a piece of storage in the cluster, provisioned by an administrator or dynamically?

- ☐ A. PersistentVolumeClaim (PVC)

- ☐ B. StorageClass
- ☐ C. PersistentVolume (PV)
- ☐ D. Volume

Correct Answer: C

Explanation: A PersistentVolume (PV) is a piece of storage provisioned for use in the cluster, independent of any individual Pod.

Domain: Container Orchestration

Competency: Storage

18. What does a PersistentVolumeClaim (PVC) represent in Kubernetes?

- ☐ A. A request for storage by a user
- ☐ B. A type of storage backend
- ☐ C. A node's local storage
- ☐ D. A backup of a volume

Correct Answer: A

Explanation: A PVC is a request made by a user (or Pod) for storage resources defined by a PV.

Domain: Container Orchestration

Competency: Storage

19. What is the function of a StorageClass in Kubernetes?

- ☐ A. To define types of storage
- ☐ B. To claim a specific PV
- ☐ C. To attach storage to a Pod
- ☐ D. To backup volume data

Correct Answer: A

Explanation: StorageClasses allow administrators to define different "classes" of storage (e.g., 'fast-ssd', 'cheap-hdd') for dynamic provisioning.

Domain: Container Orchestration

Competency: Storage

20. Which mechanism allows Kubernetes to automatically adjust the number of Pods in a Deployment based on CPU utilization or custom metrics?

- ☐ A. Vertical Pod Autoscaler (VPA)
- ☐ B. Cluster Autoscaler (CA)
- ☐ C. Horizontal Pod Autoscaler (HPA)
- ☐ D. Node Problem Detector (NPD)

Correct Answer: C

Explanation: The HPA automatically scales the number of Pod replicas

based on observed metrics like CPU or memory usage.

Domain: Cloud Native Architecture

Competency: Autoscaling

21. What is the primary goal of the Cluster Autoscaler?

- ☐ A. Scale Pod replicas
- ☐ B. Scale cluster Nodes
- ☐ C. Scale Persistent Volumes
- ☐ D. Scale Service endpoints

Correct Answer: B

Explanation: The Cluster Autoscaler adjusts the number of Nodes in the cluster based on pending Pods that cannot be scheduled due to resource limits.

Domain: Cloud Native Architecture

Competency: Autoscaling

22. What term describes an architectural approach where applications are built as small, independent services that run in their own processes?

- ☐ A. Monolithic
- ☐ B. Microservices
- ☐ C. Serverless
- ☐ D. N-Tier

Correct Answer: B

Explanation: Microservices architecture structures an application as a collection of loosely coupled, independently deployable services.

Domain: Cloud Native Architecture

Competency: Cloud Native Principles

23. What does "Serverless" computing primarily abstract away from the developer?

- ☐ A. Networking
- ☐ B. Storage
- ☐ C. Server Management
- ☐ D. Operating System choice

Correct Answer: C

Explanation: Serverless platforms (like FaaS) manage the underlying infrastructure, allowing developers to focus solely on code execution.

Domain: Cloud Native Architecture

Competency: Serverless

24. Which organization hosts Kubernetes and promotes the growth of the cloud native ecosystem?

- ☐ A. Linux Foundation (LF)
- ☐ B. Apache Software Foundation (ASF)
- ☐ C. Cloud Native Computing Foundation (CNCF)
- ☐ D. Open Source Initiative (OSI)

Correct Answer: C

Explanation: The CNCF, part of the Linux Foundation, stewards Kubernetes and many other cloud native projects.

Domain: Cloud Native Architecture

Competency: Community and Governance

25. What is the benefit of using open standards in cloud native technologies?

- ☐ A. Vendor lock-in
- ☐ B. Increased complexity
- ☐ C. Interoperability & Portability
- ☐ D. Reduced security

Correct Answer: C

Explanation: Open standards (like OCI, CRI, CNI) ensure that different tools and platforms can work together, preventing vendor lock-in.

Domain: Cloud Native Architecture

Competency: Open Standards

26. In a typical cloud native environment, which persona is primarily responsible for designing and building the application?

- ☐ A. Operator
- ☐ B. Developer
- ☐ C. Site Reliability Engineer (SRE)
- ☐ D. Security Engineer

Correct Answer: B

Explanation: Developers focus on writing, building, and testing the application code that runs within the cloud native platform.

Domain: Cloud Native Architecture

Competency: Roles and Personas

27. Which persona is typically focused on the reliability, scalability, and maintenance of the underlying Kubernetes platform?

- ☐ A. End User
- ☐ B. Application Developer
- ☐ C. Platform Operator / SRE
- ☐ D. Data Scientist

Correct Answer: C

Explanation: Operators or SREs manage the Kubernetes cluster itself, ensuring it's healthy, scalable, and available for developers.

Domain: Cloud Native Architecture

Competency: Roles and Personas

28. What are the three pillars of observability in cloud native systems?

- ☐ A. Alerts, Dashboards, Reports
- ☐ B. Logs, Metrics, Traces
- ☐ C. Monitoring, Logging, Profiling
- ☐ D. Scaling, Scheduling, Storing

Correct Answer: B

Explanation: Logs (events), Metrics (measurements over time), and Traces (request flows) are considered the fundamental pillars of observability.

Domain: Cloud Native Observability

Competency: Telemetry & Observability

29. What type of telemetry data records discrete events that happened at a specific time?

- ☐ A. Metrics
- ☐ B. Traces
- ☐ C. Logs
- ☐ D. Profiles

Correct Answer: C

Explanation: Logs provide timestamped records of events, often used for debugging or auditing.

Domain: Cloud Native Observability

Competency: Telemetry & Observability

30. What type of telemetry data represents a measurement sampled over time, often aggregated?

- ☐ A. Metrics
- ☐ B. Traces
- ☐ C. Logs
- ☐ D. Events

Correct Answer: A

Explanation: Metrics are numerical values tracked over time (e.g., CPU usage, request count) used for monitoring trends and performance.

Domain: Cloud Native Observability

Competency: Telemetry & Observability

31. What type of telemetry data shows the path and duration of a request as it flows through multiple services?

- ☐ A. Metrics
- ☐ B. Traces
- ☐ C. Logs
- ☐ D. Alerts

Correct Answer: B

Explanation: Traces (specifically distributed traces) track a request's journey across different microservices, identifying bottlenecks.

Domain: Cloud Native Observability

Competency: Telemetry & Observability

32. Which open-source monitoring system, graduated by the CNCF, is widely used for collecting and querying time-series metrics in Kubernetes?

- ☐ A. Grafana
- ☐ B. Jaeger
- ☐ C. Prometheus
- ☐ D. Fluentd

Correct Answer: C

Explanation: Prometheus is the de facto standard for metrics collection and alerting in the Kubernetes ecosystem.

Domain: Cloud Native Observability

Competency: Prometheus

33. How does Prometheus typically gather metrics from applications and infrastructure?

- ☐ A. Pushing metrics to endpoints
- ☐ B. Pulling metrics via scraping
- ☐ C. Reading log files directly
- ☐ D. Using kernel probes

Correct Answer: B

Explanation: Prometheus operates on a pull model, periodically scraping HTTP endpoints exposed by targets (applications, exporters) for metrics.

Domain: Cloud Native Observability

Competency: Prometheus

34. What language is used to write queries in Prometheus?

- ☐ A. SQL
- ☐ B. PromQL
- ☐ C. JSONPath

- ☐ D. YAML

Correct Answer: B

Explanation: PromQL (Prometheus Query Language) is the powerful functional query language used to select and aggregate time series data.

Domain: Cloud Native Observability

Competency: Prometheus

35. In cloud native cost management, what does "FinOps" primarily focus on?

- ☐ A. Optimizing application code
- ☐ B. Managing cloud spending
- ☐ C. Improving network security
- ☐ D. Automating deployments

Correct Answer: B

Explanation: FinOps is a cultural practice and framework that brings financial accountability to the variable spend model of the cloud.

Domain: Cloud Native Observability

Competency: Cost Management

36. Why can cost management be challenging in Kubernetes environments?

- ☐ A. Lack of monitoring tools
- ☐ B. Shared resources & dynamic workloads
- ☐ C. Fixed infrastructure costs
- ☐ D. Infrequent deployments

Correct Answer: B

Explanation: The shared nature of cluster resources and the dynamic scaling of workloads make it difficult to attribute costs accurately.

Domain: Cloud Native Observability

Competency: Cost Management

37. What is the practice of using Git repositories as the single source of truth for defining and managing infrastructure and applications?

- ☐ A. CI/CD
- ☐ B. DevOps
- ☐ C. GitOps
- ☐ D. Infrastructure as Code (IaC)

Correct Answer: C

Explanation: GitOps leverages Git's features (versioning, history, PRs) to manage infrastructure and application deployment declaratively.

Domain: Cloud Native Application Delivery

Competency: GitOps

38. Which core principle differentiates GitOps from general Infrastructure as Code (IaC)?

- ☐ A. Using code for infra config
- ☐ B. Automating infrastructure tests
- ☐ C. Using Git as the source of truth
- ☐ D. Manual deployment approval

Correct Answer: C

Explanation: While IaC uses code, GitOps specifically mandates Git as the declarative source of truth and uses agents to enforce that state.

Domain: Cloud Native Application Delivery

Competency: GitOps

39. What does "CI" stand for in the context of application delivery?

- ☐ A. Continuous Integration
- ☐ B. Cluster Infrastructure
- ☐ C. Container Interface
- ☐ D. Cloud Instance

Correct Answer: A

Explanation: Continuous Integration involves frequently merging code changes into a central repository, followed by automated builds and tests.

Domain: Cloud Native Application Delivery

Competency: CI/CD

40. What does "CD" stand for in CI/CD?

- ☐ A. Container Deployment
- ☐ B. Continuous Delivery/Deployment
- ☐ C. Cluster Discovery
- ☐ D. Centralized Dashboard

Correct Answer: B

Explanation: Continuous Delivery ensures code changes can be released quickly, while Continuous Deployment automatically deploys them to production.

Domain: Cloud Native Application Delivery

Competency: CI/CD

41. Which stage in a typical CI/CD pipeline for Kubernetes usually involves creating a container image?

- ☐ A. Testing
- ☐ B. Building
- ☐ C. Deployment

- ☐ D. Monitoring

Correct Answer: B

Explanation: The build stage compiles code (if necessary) and packages the application and its dependencies into a container image (e.g., Docker).

Domain: Cloud Native Application Delivery

Competency: CI/CD

42. Which tool is commonly used in CI/CD pipelines to automate the build, test, and deployment processes?

- ☐ A. Kubernetes
- ☐ B. Docker
- ☐ C. Jenkins / GitLab CI / GitHub Actions
- ☐ D. Prometheus

Correct Answer: C

Explanation: Tools like Jenkins, GitLab CI, GitHub Actions, Argo CD, Flux etc., orchestrate the steps within a CI/CD pipeline.

Domain: Cloud Native Application Delivery

Competency: CI/CD

43. What is the primary goal of Application Delivery in a cloud native context?

- ☐ A. To secure the cluster
- ☐ B. To monitor resource usage
- ☐ C. To reliably deploy & manage apps
- ☐ D. To provision infrastructure

Correct Answer: C

Explanation: Application delivery focuses on the processes and tools used to get applications running reliably and efficiently in the environment.

Domain: Cloud Native Application Delivery

Competency: Application Delivery Fundamentals

44. In Kubernetes RBAC (Role-Based Access Control), what defines a set of permissions?

- ☐ A. Role / ClusterRole
- ☐ B. ServiceAccount
- ☐ C. RoleBinding / ClusterRoleBinding
- ☐ D. User

Correct Answer: A

Explanation: A Role (namespace-scoped) or ClusterRole (cluster-scoped) contains rules that represent a set of permissions on resources.

Domain: Container Orchestration

Competency: Security

45. What object binds a Role or ClusterRole to a user, group, or ServiceAccount?

- ☐ A. Secret
- ☐ B. ConfigMap
- ☐ C. NetworkPolicy
- ☐ D. RoleBinding / ClusterRoleBinding

Correct Answer: D

Explanation: A RoleBinding (namespace-scoped) or ClusterRoleBinding (cluster-scoped) grants the permissions defined in a Role/ClusterRole to subjects.

Domain: Container Orchestration

Competency: Security

46. If you need a Pod to run on every single Node in the cluster (or a subset), which controller is most suitable?

- ☐ A. Deployment
- ☐ B. StatefulSet
- ☐ C. DaemonSet
- ☐ D. Job

Correct Answer: C

Explanation: DaemonSets ensure that all (or some) Nodes run a copy of a Pod, useful for cluster services like log collectors or node monitors.

Domain: Kubernetes Fundamentals

Competency: Kubernetes Resources

47. What is a primary difference between a Deployment and a StatefulSet?

- ☐ A. StatefulSets manage Pods
- ☐ B. Deployments are for databases
- ☐ C. StatefulSets provide stable IDs
- ☐ D. Deployments use PVs

Correct Answer: C

Explanation: StatefulSets provide guarantees about the ordering and uniqueness of Pods, including stable network identifiers and persistent storage.

Domain: Kubernetes Fundamentals

Competency: Kubernetes Resources

48. Which Kubernetes API object is used to manage external access to services in a cluster, typically HTTP/S?

- ☐ A. Service (LoadBalancer type)

- ☐ B. NodePort Service
- ☐ C. Ingress
- ☐ D. ExternalName Service

Correct Answer: C

Explanation: Ingress provides HTTP/S routing rules to manage external users' access to services within the cluster based on hostnames or paths.

Domain: Container Orchestration

Competency: Networking

49. What design pattern involves running an additional container within a Pod to provide auxiliary functions (like logging or monitoring) to the main application container?

- ☐ A. Init Container
- ☐ B. Sidecar Container
- ☐ C. Ephemeral Container
- ☐ D. Job Container

Correct Answer: B

Explanation: The Sidecar pattern adds helper containers alongside the main application container within the same Pod network/storage namespace.

Domain: Kubernetes Fundamentals

Competency: Containers

50. What fundamental concept allows Kubernetes to maintain the desired state declared by the user?

- ☐ A. Imperative Commands
- ☐ B. Control Loop / Reconciliation
- ☐ C. Manual Scaling
- ☐ D. Direct Node Access

Correct Answer: B

Explanation: Kubernetes controllers operate on control loops, constantly comparing the desired state (from API objects) with the actual cluster state.

Domain: Kubernetes Fundamentals

Competency: Kubernetes Architecture